

Draft CalEnviroScreen 5.0 Results Data Dictionary and Information on Zeros and Missing Values

This document accompanies the Draft CalEnviroScreen 5.0 results available as an excel spreadsheet. Draft CalEnviroScreen 5.0 was released January 2026.

The Draft CalEnviroScreen 5.0 results are available to view in the [online map](#).

For more information and to access additional data, see the [Draft CalEnviroScreen 5.0 webpage](#).

Data Dictionary for Results

Data Dictionary for Draft CES 5.0 Results Data Spreadsheet	
Variable Name	Description
Census Tract	Census Tract ID from 2020 Decennial Census
Total Population	2023 5-year ACS population estimates in census tracts
California County	California county that the census tract falls within
ZIP	Postal ZIP Code that the census tract falls within
Approximate Location	Approximate city or unincorporated area where each census tract is located based on Esri Demographics data (Esri, 2024). This is for reference purposes only and should not be used to determine whether a census tract falls within a city or town boundary.
Draft CES 5.0 Score	CalEnviroScreen Score, Pollution Score multiplied by Population Characteristics Score
Draft CES 5.0 Percentile	Percentile of the CalEnviroScreen Score
Draft CES 5.0 Percentile Range	Percentile of the CalEnviroScreen Score, grouped by 5% increments
Ozone	Mean of summer months (May - Oct.) of the daily maximum 8-hour ozone concentration
Ozone Pctl	Ozone percentile
PM2.5	Annual mean PM2.5 concentrations

PM2.5 Pctl	PM2.5 percentile
Child's Lead	Potential risk for lead exposure in children living in low-income communities with older housing
Child's Lead Pctl	Children's Lead Risk from Housing percentile
Diesel PM	Diesel PM emissions from on-road and non-road sources within and near populated blocks
Diesel PM Pctl	Diesel PM percentile
Drinking Water	Drinking water contaminant index for selected contaminants
Drinking Water Pctl	Drinking Water Contaminants percentile
Pesticides	Total pounds of selected active pesticide ingredients (filtered for hazard and volatility) used in production-agriculture per square mile in the census tract
Pesticides Pctl	Pesticide Use percentile
Tox. Release	Toxicity-weighted air concentrations from chemical releases by large facilities
Tox. Release Pctl	Toxic Releases from Facilities percentile
Traffic	Traffic density, in vehicle-kilometers per hour per road length, within 150 meters of the census tract boundary
Traffic Pctl	Traffic Impacts percentile
Cleanup Sites	Sum of weighted EnviroStor cleanup sites and National Priorities List Superfund sites within buffered distances to populated blocks of census tracts
Cleanup Sites Pctl	Cleanup Sites percentile
Groundwater Threats	Sum of weighted GeoTracker cleanup, land disposal, underground storage tank, and produced water pond sites and dairies and feedlots sites within buffered distances to populated blocks of census tracts
Groundwater Threats Pctl	Groundwater Threats percentile
Haz. Waste	Sum of weighted hazardous waste facilities and large quantity generators within buffered distances to populated blocks of census tracts
Haz. Waste Pctl	Hazardous Waste percentile

Impaired Waters	Sum of number of pollutants across all impaired water bodies within buffered distances to populated blocks and census tracts
Impaired Waters Pctl	Impaired Waters percentile
Sm. Air Tox. Sites	Sum of CEIDARS emissions sites and oil and natural gas well sites within buffered distances to populated blocks of census tracts
Sm. Air Tox. Sites Pctl	Small Air Toxic Sites percentile
Solid Waste	Sum of weighted solid waste sites and facilities (SWIS) within buffered distances to populated blocks of census tracts
Solid Waste Pctl	Solid Waste percentile
Pollution Burden	Average of percentiles from the Pollution Burden indicators (with a half weighting for the Environmental Effects indicators)
Pollution Burden Score	Pollution Burden variable scaled with a range of 0-10. (Used to calculate draft CES 5.0 Score)
Pollution Burden Pctl	Pollution Burden percentile
Asthma	Age-adjusted rate of emergency department visits for asthma
Asthma Pctl	Asthma percentile
Cardiovascular Disease	Age-adjusted rate of emergency department visits for heart attacks per 10,000
Cardiovascular Disease Pctl	Cardiovascular Disease percentile
Diabetes	Model-based prevalence of diabetes in adults, age 18 years or older
Diabetes Pctl	Diabetes Prevalence percentile
Low-Birth-Weight	Percent of infants weighing less than 2,500gram (~5.5 pounds)
Low-Birth-Weight Pctl	Low-birth-weight Infants percentile
Education	Percentage of population over 25 with less than a high school education
Education Pctl	Educational Attainment percentile
Housing Burden	Percentage of households that are both low-income and severely burdened by housing costs
Housing Burden Pctl	Housing Burden percentile

Linguistic Isolation	Percentage of households where no one over age 14 speaks English well
Linguistic Isolation Pctl	Linguistic Isolation percentile
Poverty	Percentage of people living below twice the federal poverty level
Poverty Pctl	Poverty percentile
Unemployment	Percentage of people aged 16 and older who are unemployed and eligible for the workforce.
Unemployment Pctl	Unemployment percentile
Pop. Char.	Average of percentiles from the Population Characteristics indicators
Pop. Char. Score	Population Characteristics variable scaled with a range of 0-10. (Used to calculate draft CES 5.0 Score)
Pop. Char. Pctl	Population Characteristics percentile

Data Dictionary for Demographic Profile

Data Dictionary for Demographic Profile Tab	
Variable Name	Description
Census Tract	Census Tract ID from 2020 Decennial Census
Draft CES 5.0 Score	CalEnviroScreen Score, Pollution Score multiplied by Population Characteristics Score
Draft CES 5.0 Percentile	Percentile of the CalEnviroScreen Score
Draft CES 5.0 Percentile Range	Percentile of the CalEnviroScreen Score, grouped by 5% increments
California County	California county where the census tract falls within
Total Population	2023 5-year ACS population estimates in census tracts
Children < 10 years (%)	2023 5-year ACS population estimates of the percent per census tract of children under 10 years old
Pop 10-64 years (%)	2023 5-year ACS population estimates of the percent per census tract of people between 10 and 64 years old
Elderly > 64 years (%)	2023 5-year ACS population estimates of the percent per census tract of elderly 65 years and older

Hispanic (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as Hispanic or Latino
White (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as non-Hispanic white
Black (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as non-Hispanic African American or black
Native American (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as non-Hispanic Native American
Asian American (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as non-Hispanic Asian or Pacific Islander
Other/Multiple (%)	2023 5-year ACS population estimates of the percent per census tract of those who identify as non-Hispanic "other" or as multiple races

Description of Zeros and Missing Values

This summarizes the zero and NA values from the draft CalEnviroScreen 5.0 results excel spreadsheet. 9,106 California census tracts are used in the CalEnviroScreen analysis.

Pollution Burden Indicators

Exposure Indicators

Indicator	Ozone	PM2.5	Child's Lead	Diesel PM	Drinking Water	Pesticide Use	Toxic Release	Traffic
# of Zeros	0	0	10	0	0	5980	37	11
# of NAs	0	0	89	9	9	0	0	0

Environmental Effects Indicators

Indicator	Cleanup Sites	Groundwater Threats	Hazardous Waste	Impaired Waters	Small Air Toxic Sites	Solid Waste
# of Zeros	2328	2172	855	3349	726	4138
# of NAs	0	0	0	0	0	0

Population Characteristics Indicators

Sensitive Population Indicators

Indicator	Asthma	Cardiovascular Disease	Diabetes Prevalence	Low-Birth-Weight Infants
# of Zeros	0	0	0	13
# of NAs	33	26	36	234

Socioeconomic Factors Indicators

Indicator	Education	Housing Burden	Linguistic Isolation	Poverty	Unemployment
# of Zeros	84	32	685	12	93
# of NAs	29	77	60	53	48

What do Indicator Raw Values of Zero and NA Mean?

NA or Missing Raw values:

When an indicator has a missing value (“NA”), it usually means that no monitoring or reporting was conducted, no population was reported, or the estimates for that indicator were considered unreliable within that census tract. For example, Low-Birth-Weight Infants data are not available for all California census tracts because some areas have very small populations or too few registered births to calculate a reliable score. Additionally, 25 out of 9,106 census tracts reported zero population in the 2019–2023 ACS. These tracts were assigned “NA” for all Population Characteristic indicators. The Low-Birth-Weight Infants and Diabetes Prevalence indicators require at least 50 data points per census tract (births and adults over age 18, respectively) over the analyzed period. If this threshold is not met, the indicator is assigned “NA” for that tract.

Zero raw values:

A raw value of zero typically means monitoring or reporting occurred, but no impacts were found. For many Exposure and Environmental Effects indicators, this means no facilities or sites were located within 1,000 meters of a populated area in the census tract. For example, 2,328 census tracts did not have a cleanup site within 1,000 meters of a populated area; these tracts were given a value of zero. Similarly, a zero for a Population Characteristic indicator such as Poverty means people live there, but none are below the poverty level.

What do Percentiles of Zero and NA Mean?

NA or Missing Percentile Values:

The indicators with missing values (“NA”) are Children’s Lead Risk from Housing, Diesel PM, Drinking Water Contaminants, and all Population Characteristic indicators. These indicators are assigned no percentile (shown as “NA”) and do not contribute to the overall CalEnviroScreen score. For example, if a census tract is missing both Children’s Lead Risk from Housing and Drinking Water Contaminants scores, the denominator for Exposure indicators is adjusted to six instead of eight.

Zero Percentile Values

Many census tracts have a raw value of zero for Exposure and Environmental Effects indicators. These tracts are excluded from percentile calculations to avoid falsely suggesting an impact. For instance, about 66% of census tracts have none of the selected pesticides. Including these zeros in percentile calculations would incorrectly place them at the 66th percentile. Instead, these tracts are assigned a percentile value of “0,” indicating no impact. For the Pesticide Use indicator, 3,126 census tracts (9,106 minus 5,980) make up the percentile range. Indicators with missing values (“NA”) are never included in percentile calculations. For example, the Educational Attainment indicator includes 8,993 census tracts in its percentile calculation (9,106 minus 84 zero values minus 29 missing values = 8,993).

Impact of Zero and Nas on CalEnviroScreen Scoring

Census tracts with missing estimates for more than four Population Characteristics indicators (i.e., more than half) are assigned an “NA” for both the Population Characteristics component score and the overall CalEnviroScreen score. This approach prevents over-reliance on a small number of indicators and maintains the emphasis on cumulative burden. Because there are fewer “NA” values within the Pollution Burden component, all census tracts in the state receive a Pollution Burden score.

Additionally:

- Census tracts with fewer than 50 people receive an “NA” for Population Characteristics.
- Census tracts with more than 50 people, but fewer than 50 living outside of group quarters, also receive an “NA” for Population Characteristics. (Group quarters include places such as college dormitories, prisons, and residential treatment centers.)